

Cover Page

Jubail Industrial College

Department of Computer and Information Technology



LifeLink

“Connecting Lives Through Blood”

Project report

Course Code: CS 464

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Abstract

Blood shortages, low public awareness, and poor coordination between donors and healthcare professionals are among the difficulties that many hospitals encounter despite the fact that blood donation is a vital healthcare service. This project presents LifeLink, a smartphone software that uses a single digital platform to link hospitals, patients, and donors. For users of all ages, the program prioritizes usability and simplicity to guarantee simple navigation and efficient communication. Features including blood request management, donor notifications, donation history tracking, and an interactive prototype created with Figma are all part of LifeLink. LifeLink seeks to enhance healthcare services, promote blood donation, and lessen delays in emergency circumstances by enhancing coordination and user experience.

Table of Contents

Cover Page	1
Acknowledgment	2
Abstract	3
Introduction	5
Summary of Articles	6
Flowchart	9
Sketches	10
Prototype	11
Conclusion.....	13
References	14

Introduction

Blood donation — a life-saving health care service — enables all sorts of medical procedures, from emergency care and surgeries to treating chronic diseases. Hospitals and health care providers need a safe and adequate supply of blood to meet patients' needs. However, access, public awareness, and poor coordination between donors and health organizations are some of the most obvious issues that have made the blood donation process extremely challenging. These challenges lead to delayed blood availability and low participation in blood donation campaigns.

Digital technologies and health care services recently became increasingly relevant in terms of improving patient care, quality of care provision, and inter-communication among the various health care staff in the health care delivery system. It has especially enhanced information that can be accessed more easily, organized information, and contact with customers through applications. But these apps also need to be convincing in usability, clarity, and needs.

LifeLink aims to rectify this with everything listed above in a single integrated digital package bringing patients, hospitals, and donors as one friendly user environment. The app is designed to communicate with people of all ages to better understand and help you donate blood to blood donors, which will also be followed with information, navigation, and the kind of content that will save lives. This study focuses on the application design and application implementation of the LifeLink application, which uses technology to engage local stakeholder groups, enhancing the blood donation process, delivering blood donation experience, and facilitating saving opportunities.

Summary of Articles

Alessa, T., Hawari, A. I., & Alshahrani, A. (2022)

Summary: The Wateen app is a national digital platform developed in collaboration with the Saudi Ministry of Health to enhance the blood donation process. Its main objective is to ease blood donation and improve access to it by facilitating communication between donors and hospitals/blood banks. This app can be used to register as a donor, book an appointment for donation, an overview of the donation history and notifications of blood shortages or urgent requirements. The ultimate aim of the app is to encourage voluntary donation as opposed to replacement or emergency donations and supporting the vision of Saudi Arabia presented in 2030 namely to improve the digital health of citizens and participation of it. This mixed methodology consisted of assessing the Wateen App through the execution of an online questionnaire (401 donors) and semi-structured interviews with healthcare professionals. The app was found to be very usable and acceptable, with specific utilities such as booking appointments, a digital donor card and notifications. However, there were also complaints with regard to ease of use (size of fonts, language, older people) and limited extent of features and function.

Li, L. et al. (2023)

Summary:

How mobile applications help the hospitals and donors and make blood donation much easier. It's also reviews several other studies about blood donation apps 952 young adult to see what they thought about the apps. It's a big problem because many hospitals are running low on blood, and others forget to donate or don't know where to go, Lee said. Those apps come in handy by directing you to the closest places to donate blood, reminding you when it's time to give again, and enabling you easily make an appointment.

Chamorro, R.E. et al. (2024)

Summary:

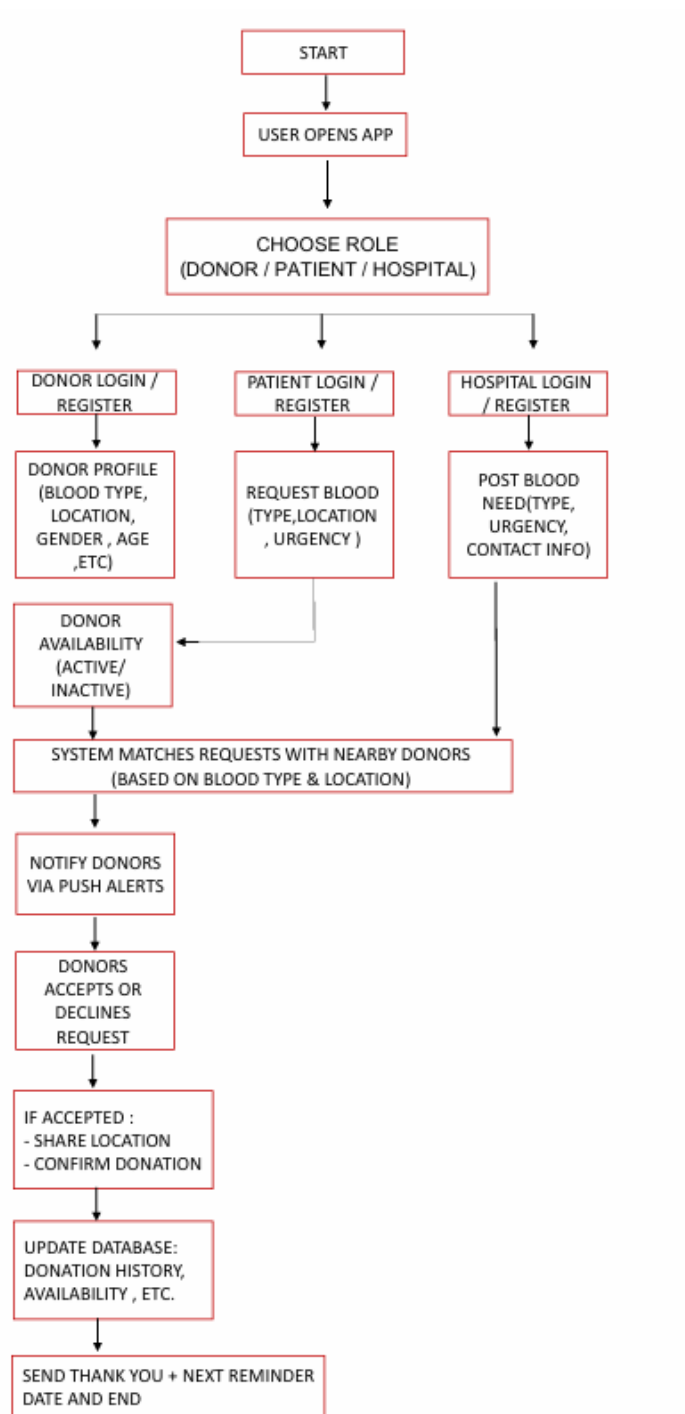
In this paper we investigated how gamification can assist the deferred to pursue blood donation--a category which is generally neglected in other donor engagement systems. The authors modelled the prototype of a gamified application, working with three key motivators: awareness and knowledge, interaction and validation, and motivation. Guided by MDA, the prototype is developed (based on formative work) and trialed with 11 participants evaluating usability and perceived utility. The results revealed good usability (SUS = 70.9/100) and overall positive feedback regarding the idea. The study found that gamified platforms returned deferrers' feelings of reward and could help maintain donors' motivation, with more testing required.

Mostafa, A.M., Youssef, A.E., & Alshorbagy, G. (2014)

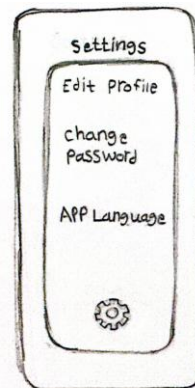
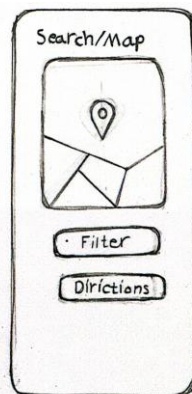
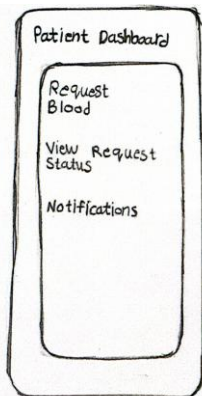
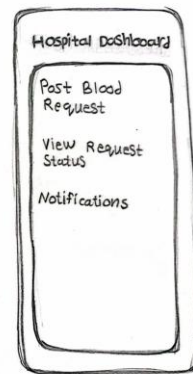
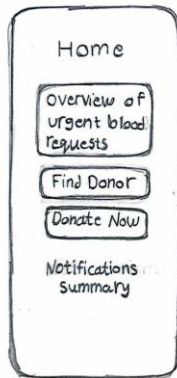
Summary:

This study explains a smart system made to make blood donation easier and quicker. It depends on cloud computing and mobile apps to link donors with hospitals and blood banks. The main goal is to make registration and communication simple, especially when blood is urgently needed. The system also uses social media to raise awareness and motivate people to donate. According to the authors, using this system widely could improve how blood donation works, increase the number of regular donors, and reduce blood shortages in hospitals.

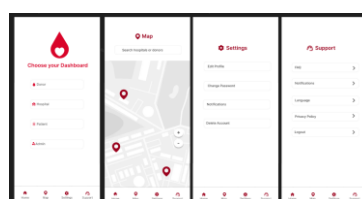
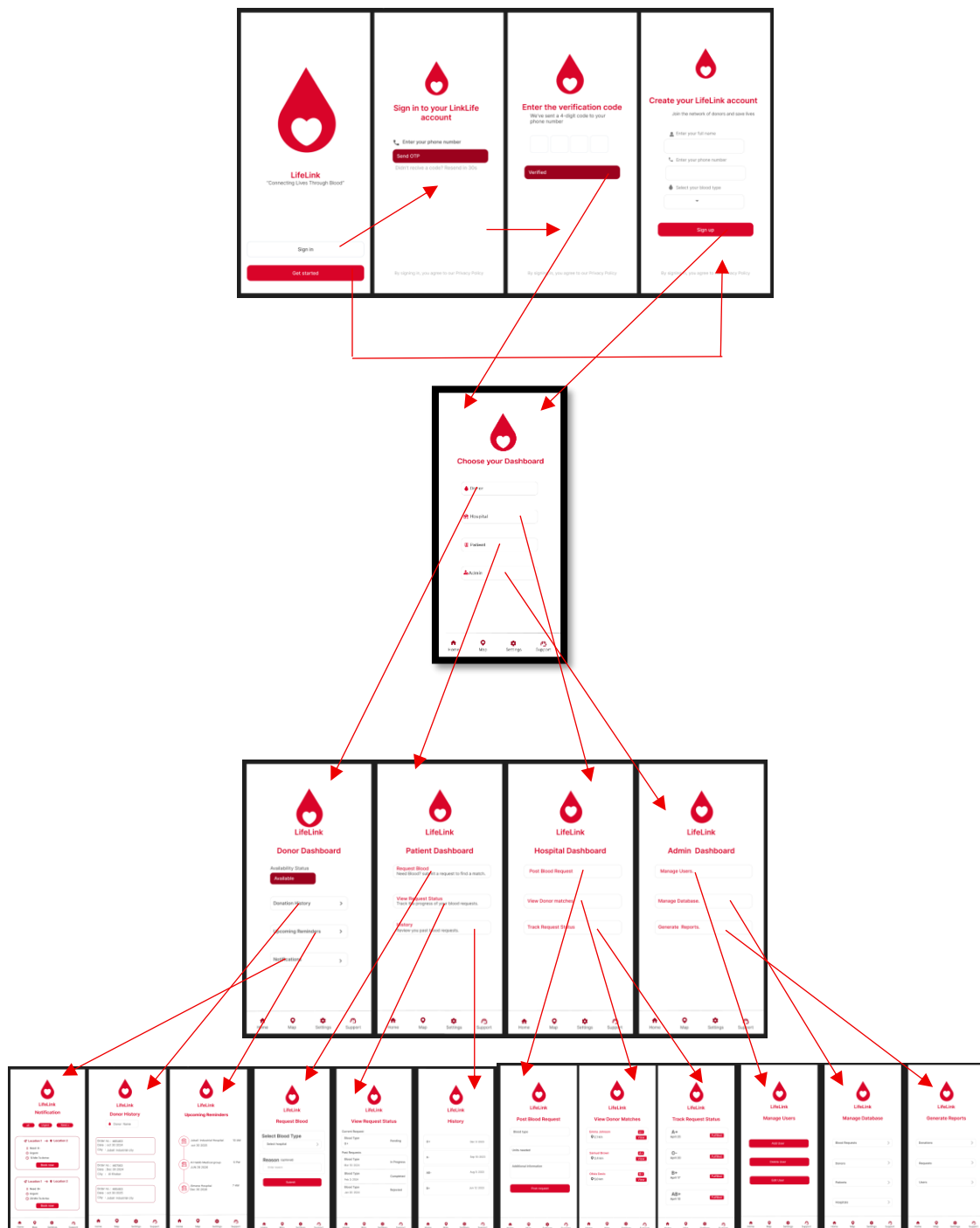
Flowchart



Sketches



Prototype



These are shared pages it can be accessed from the navigation bar on the pages

The link:

<https://www.figma.com/design/PHGVVkNrr8M0BpCMtAXHhq/Untitled?node-id=0-1&t=MIhxf60NSacpBk-1>

Conclusion

This project focused on designing LifeLink, a blood donation application that aims to make the process of requesting and donating blood clearer, faster, and more accessible for everyone involved. By bringing patients, donors, and hospitals into one platform, the application helps reduce confusion and improve communication during critical situations.

The design of the application emphasizes simplicity and ease of use, allowing users to navigate the system, submit blood requests, track request status, and view donation history without difficulty. These features support users in urgent moments where time and clarity are essential.

Overall, LifeLink demonstrates how a well-designed digital solution can support health care services and encourage community participation in blood donation. By improving awareness, coordination, and user experience, the application contributes to creating a more efficient and reliable blood donation process that ultimately helps save lives. At its core, LifeLink reflects the importance of using technology responsibly to support human needs and strengthen collaboration in life-saving health services.

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